Testing Dynamical Dark-Energy Model with Standard Sirens

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Bright standard sirens from the Einstein Telescope (ET) provide an independent probe of cosmic expansion. In this article, we test whether future ET observations can constrain a dynamical dark-energy scenario motivated by DESI DR2. Using the best-fitting 29-bin dark-energy equation of state as the fiducial cosmology, we generate mock catalogs of 100, 500, 1000, and 3000 binary-neutron-star standard sirens with known host redshifts, and reconstruct the equation of state using both GW-only data and combined analyses with DESI DR2 BAO, BBN, and a compressed CMB constraint. The effective numbers of constrained modes are $N_{\rm eff} \sim 4-5$ (GW-only) and $5-6$ (combined), with information concentrated in a few correlated modes. The GW3000 catalog yields a departure from ΛCDM exceeding 5σ, and combined analyses reach similar significance for $\mathcal{O}(10^3)$ events. Comparing flat ΛCDM with its Chevallier–Polarski–Linder extension using GW data alone gives significances of 3.3σ and 6.3σ for 1000 and 3000 events, rising to 7.7σ and 11.0σ with external data. These results indicate that a few thousand ET bright standard sirens could provide a sensitive test of departures from ΛCDM model.

I. INTRODUCTION

The discovery of the accelerated expansion of the Universe established dark energy as one of the central questions in modern cosmology [1, 2]. The spatially flat ΛCDM model has provided a remarkably successful description of a wide range of observations, including measurements of the cosmic microwave background (CMB) and baryon acoustic oscillations (BAO) [3]. Nevertheless, the physical origin of late-time cosmic acceleration remains unknown. A key observational question is whether the expansion history is fully consistent with a cosmological constant or whether it contains signatures of a dynamical dark-energy component.

Recent BAO measurements from the Dark Energy Spectroscopic Instrument (DESI) Data Release 2 (DR2) have renewed interest in this question [5, 6]. By combining DESI DR2 BAO measurements with Big Bang nucleosynthesis (BBN), compressed CMB acoustic-scale constraints, and Type Ia supernovae (SNe Ia), Ref. [7] reconstructed the dark-energy equation of state in a non-parametric manner. In this analysis, the dark-energy history was represented by 29 redshift bins over $0 < z \leq 2.5$, with correlations between neighboring bins regularized through a Horndeski-motivated prior [8]. The resulting reconstruction exhibits a redshift-dependent deviation from ΛCDM, providing a possible indication of dynamical dark energy.

However, interpreting such a deviation requires independent confirmation from complementary probes. In particular, the late-time distance information entering current analyses relies significantly on SNe Ia. Although SNe Ia have played a fundamental role in establishing cosmic acceleration, their cosmological application involves empirical standardization relations based on light-curve properties and spectral information [9, 10]. The absolute magnitude calibration is usually marginalized as a nuisance parameter, while residual uncertainties associated with photometric calibration, light-curve modeling, and survey systematics may affect the inferred distance–redshift relation [11, 12]. Furthermore, standardized SNe Ia luminosities exhibit correlations with host-galaxy properties, introducing additional astrophysical uncertainties in precision cosmology [13]. Possible population evolution of SNe Ia with redshift may also introduce systematic effects in the interpretation of their cosmological distances [14].

Gravitational-wave (GW) standard sirens provide an independent route to measure the expansion history with different systematic uncertainties [15, 16]. Unlike electromagnetic distance indicators, the luminosity distance of a standard siren is directly encoded in the GW waveform. When the source redshift is obtained from an electromagnetic counterpart or host-galaxy identification, bright standard sirens provide a direct measurement of the luminosity distance–redshift relation. The observation of GW170817 demonstrated this approach and established the foundation for GW-based cosmological measurements [17, 18].

Future third-generation detectors, such as the Einstein Telescope (ET), are expected to detect large populations of binary-neutron-star mergers and may provide thousands of bright standard sirens extending to higher redshifts [19–21]. Such a large and independent distance sample offers a unique opportunity to test whether the apparent departure from ΛCDM inferred from current electromagnetic observations can be reproduced by a fundamentally different cosmological probe.

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In this work, we investigate the capability of ET bright standard sirens to test the DESI DR2-inspired dynamical dark-energy history. We adopt its best-fitting 29-bin $w(z)$ reconstruction as a fiducial scenario and ask whether future GW observations can recover and distinguish this expansion history from ΛCDM. We generate mock catalogs of $N = 100$, 500, 1000, and 3000 binary-neutron-star bright standard sirens with known redshifts, including instrumental distance uncertainties and weak-lensing scatter.

We perform two complementary analyses. First, combining ET catalogs with DESI DR2 BAO, BBN, and compressed CMB constraints, we reconstruct the full 29-dimensional $w(z)$ history. Principal component analysis (PCA) is then used to identify the redshift-dependent combinations constrained by the data. This allows us to quantify the effective dimensionality of the measurable dark-energy space and evaluate the covariance-weighted deviation from ΛCDM. Second, we analyze the GW catalogs independently within a low-dimensional framework by fitting flat ΛCDM and the Chevallier–Polarski–Linder (CPL) parameterization [22, 23]. Since the fiducial history is intrinsically nonparametric, the inferred (w_0, w_a) parameters should be interpreted as an effective projection of the underlying evolution rather than a complete description of the reconstructed $w(z)$ behavior. This analysis provides a complementary assessment of whether ET standard sirens can identify a departure from ΛCDM using a commonly adopted dark-energy parameterization.

Together, these two approaches address different aspects of future GW cosmology: the nonparametric analysis determines which redshift-dependent features of the dark-energy history are observationally accessible, while the CPL analysis quantifies the detectability of the deviation within a low-dimensional phenomenological model. The central goal of this work is therefore to assess whether the DESI DR2-motivated deviation from ΛCDM can be independently tested by future standard-siren observations, how the sensitivity scales with the size of the GW catalog, and which combinations of $w(z)$ contain the dominant discriminating information.

The paper is organized as follows. Sec. II introduces the cosmological framework, while Sec. III describes the GW signal and ET response. Sec. IV presents the generation of mock catalogs. Sec. V describes the external cosmological datasets. Sec. VI presents the inference methodology, including nonparametric reconstruction of $w(z)$, principal component analysis, and the parametric CPL analysis. We summarize our conclusions in Sec. VII.

II. DARK-ENERGY COSMOLOGICAL FRAMEWORK

We describe the background geometry by the Friedmann–Lemaître–Robertson–Walker (FLRW) metric,

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a^2(t)\left[\frac{\mathrm{d}r^2}{1-kr^2} + r^2\left(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2\right)\right], \tag{1}$$

where t is cosmic time, (r, θ, ϕ) are comoving spatial coordinates, and $a(t)$ is the scale factor. The values $k = 0$, 1, and -1 describe flat, closed, and open geometries, respectively. We restrict the analysis to $k = 0$ in this article.

In general relativity, with minimally coupled radiation, baryons, cold dark matter, and dark energy, the background expansion obeys

$$H^2 = \frac{8\pi G}{3}\rho_{\rm tot}, \tag{2}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho_{\rm tot} + 3p_{\rm tot}\right), \tag{3}$$

where $H \equiv \dot{a}/a$, an overdot denotes differentiation with respect to t, and G is Newton's gravitational constant. The total energy density and pressure are

$$\rho_{\rm tot} = \rho_r + \rho_b + \rho_c + \rho_{\rm de}, \qquad p_{\rm tot} = p_r + p_b + p_c + p_{\rm de}, \tag{4}$$

where the subscripts r, b, c, and de denote radiation, baryons, cold dark matter, and dark energy. We also define $\rho_m \equiv \rho_b + \rho_c$ and $p_m \equiv p_b + p_c$. Radiation is negligible at the late times relevant here. Because the components interact only gravitationally, each obeys the continuity equation

$$\dot{\rho}_i + 3H\left(1 + w_i\right)\rho_i = 0. \tag{5}$$

The equation-of-state parameter is $w_i \equiv p_i/\rho_i$. Matter has $w = 0$, whereas dark energy can have a redshift-dependent equation of state, $w = w(z)$. For instance, in ΛCDM, dark energy is a cosmological constant with

$$w(z) = -1. \tag{6}$$

In the CPL model, its equation of state is [22, 23]

$$w(z) = w_0 + w_a\frac{z}{1+z}, \tag{7}$$

where w_0 is the present-day value and w_a controls the redshift evolution. The choice $(w_0, w_a) = (-1, 0)$ recovers ΛCDM.

Using $a = (1+z)^{-1}$, the dark-energy density evolves as

$$\rho_{\rm de}(z) = \rho_{\rm de,0}\exp\left[3\int_0^z \frac{1 + w(z')}{1+z'}\,\mathrm{d}z'\right]. \tag{8}$$

Substituting the total energy density into Eq. (2) and neglecting late-time radiation gives

$$H(z) = H_0\left[\Omega_m(1+z)^3 + \Omega_{\rm de}\frac{\rho_{\rm de}(z)}{\rho_{\rm de,0}}\right]^{1/2}. \tag{9}$$

Here Ω_m and $\Omega_{\rm de}$ are the present matter and dark-energy density parameters, with $\Omega_{\rm de} = 1-\Omega_m$ for a flat universe. The luminosity distance is

$$d_L(z) = (1+z)d_C(z), \tag{10}$$

where

$$d_C(z) = c\int_0^z \frac{{\rm d}z'}{H(z')}. \tag{11}$$

III. GRAVITATIONAL-WAVE SIGNAL AND EINSTEIN TELESCOPE

Unlike standard candles, GWs from inspiraling compact binaries provide an absolute measurement of luminosity distance [15, 16]. For a binary neutron star (BNS), the phase evolution constrains the chirp mass, while the waveform amplitude encodes the luminosity distance [24]. BNS mergers can therefore serve as standard sirens. We use geometrized units, $G = c = 1$, in the waveform and detector-response calculations.

Ground-based interferometers measure the dimensionless strain induced by a passing GW. In the transverse-traceless (TT) gauge, the response is

$$h(t) = F_+(\theta,\phi,\psi)\,h_+(t) + F_\times(\theta,\phi,\psi)\,h_\times(t), \tag{12}$$

where the independent polarizations satisfy $h_+ = h_{xx} = -h_{yy}$ and $h_\times = h_{xy} = h_{yx}$. The functions F_+ and $F_\times$ are the detector antenna patterns, ψ is the polarization angle, and (θ,ϕ) gives the source position in the detector frame [31].

ET consists of three 60° interferometers arranged in an equilateral triangle and rotated by 120° with respect to one another. Their antenna patterns are [19]

$$\begin{aligned}
F_+^{(1)}(\theta,\phi,\psi) =& \frac{\sqrt{3}}{2}\left[\frac{1}{2}\left(1+\cos^2(\theta)\right)\cos(2\phi)\cos(2\psi)\right.\\
&\left.-\cos(\theta)\sin(2\phi)\sin(2\psi)\right]\\
F_\times^{(1)}(\theta,\phi,\psi) =& \frac{\sqrt{3}}{2}\left[\frac{1}{2}\left(1+\cos^2(\theta)\right)\cos(2\phi)\sin(2\psi)\right.\\
&\left.+\cos(\theta)\sin(2\phi)\cos(2\psi)\right]\\
F_{+,\times}^{(2)}(\theta,\phi,\psi) =& F_{+,\times}^{(1)}(\theta,\phi+2\pi/3,\psi)\\
F_{+,\times}^{(3)}(\theta,\phi,\psi) =& F_{+,\times}^{(1)}(\theta,\phi+4\pi/3,\psi).
\end{aligned} \tag{13}$$

We model nonspinning, quasicircular BNS signals with the TaylorF2 frequency-domain waveform in the stationary-phase and restricted post-Newtonian approximations [26]. The amplitude is kept at leading Newtonian order, proportional to $f^{-7/6}$, and the phase is computed through 3.5 post-Newtonian order. The Fourier transform $\mathcal{H}(f)$ of $h(t)$ is

$$\begin{aligned}
\mathcal{H}(f) =& \mathcal{A}f^{-7/6}\exp\left\{i\left[2\pi f t_0 - \frac{\pi}{4} + 2\Psi\left(\frac{f}{2}\right)\right.\right.\\
&\left.\left.-\varphi_{(2,0)}\right]\right\},
\end{aligned} \tag{14}$$

where the Fourier amplitude $\mathcal{A}$ is

$$\begin{aligned}
\mathcal{A} =& \frac{1}{d_L}\left[F_+^2\left(1+\cos^2\iota\right)^2 + 4F_\times^2\cos^2\iota\right]^{1/2}\\
&\times\sqrt{\frac{5\pi}{96}}\,\pi^{-7/6}\mathcal{M}_{c,\rm obs}^{5/6}.
\end{aligned} \tag{15}$$

Here t_0 is the coalescence time, and ι is the angle between the orbital angular momentum and the line of sight. For component masses m_1 and m_2, the total mass is $M = m_1 + m_2$, the symmetric mass ratio is $\eta = m_1 m_2/M^2$, and the source-frame chirp mass is $\mathcal{M}_c = M\eta^{3/5}$. The detector-frame chirp mass is $\mathcal{M}_{c,\rm obs} = (1+z)\mathcal{M}_c$. The functions $\Psi(f)$ and $\varphi_{(2,0)}$ describe the post-Newtonian phase evolution and the polarization-dependent phase offset of the dominant harmonic, respectively [19, 24].

IV. MOCK GW CATALOGS

We construct the fiducial expansion history from the nonparametric reconstruction of $w(z)$ in Ref. [7]. That reconstruction used DESI DR2 BAO measurements, a BBN prior [27], and a compressed Planck Public Release 4 (PR4) constraint on the CMB acoustic scale [28]. It represented the equation of state by 29 piecewise-constant bins, uniformly spaced in scale factor over $0 < z \le 2.5$, and imposed a Horndeski-motivated correlation prior [8]. We use the released best-fitting amplitudes w_i as the fiducial history:

$$w_{\rm fid}(z) = \begin{cases} w_i, & z_{i,\min} < z \le z_{i,\max}, \quad i = 1,\ldots,29,\\ -1, & z > 2.5, \end{cases} \tag{16}$$

where $z_{i,\min}$ and $z_{i,\max}$ are the boundaries of bin i. Setting $w_{\rm fid} = -1$ at $z > 2.5$ preserves the standard early-Universe expansion history.

For the luminosity-distance calculation, this prescription amounts to the discretization

$$w(z) \to \{w_1, w_2, \ldots, w_{29}\}. \tag{17}$$

This piecewise-constant form permits an analytic evaluation of the dark-energy density in Eq. (8) within each bin. For $w(z) = w_i$ over $z \in [z_i, z_{i+1}]$,

$$\int_{z_i}^{z}\frac{1+w_i}{1+z'}\,{\rm d}z' = (1+w_i)\ln\frac{1+z}{1+z_i}, \tag{18}$$

which determines the binned evolution of $\rho_{\rm de}(z)$ and hence $H(z)$.

Figure 1 shows the adopted history and its uncertainty bands. Table I lists the best-fitting bin values used to calculate luminosity distances.

To ensure consistency between the fiducial $w(z)$ history and the background cosmology, we adopt the background parameters from the same best-fit reconstruction: $H_0 = 57.365\ {\rm km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_m = 0.423$ [7]. We evaluate the dark-energy density with Eq. (18) and obtain

TABLE I. Best-fitting equation-of-state amplitudes for the fiducial DESI DR2 history [7]. Within bin i, $w_{\rm fid}(z) = w_i$.

i	w_i	i	w_i
1	0.8320	16	-0.8635
2	0.7260	17	-1.1272
3	0.5847	18	-1.3825
4	0.3166	19	-1.5248
5	-0.0088	20	-1.5153
6	-0.3689	21	-1.3809
7	-0.7129	22	-1.1999
8	-0.9973	23	-1.0302
9	-1.1787	24	-0.9123
10	-1.2336	25	-0.8460
11	-1.1538	26	-0.8225
12	-0.9813	27	-0.8234
13	-0.7915	28	-0.8308
14	-0.6726	29	-0.8371
15	-0.6911		

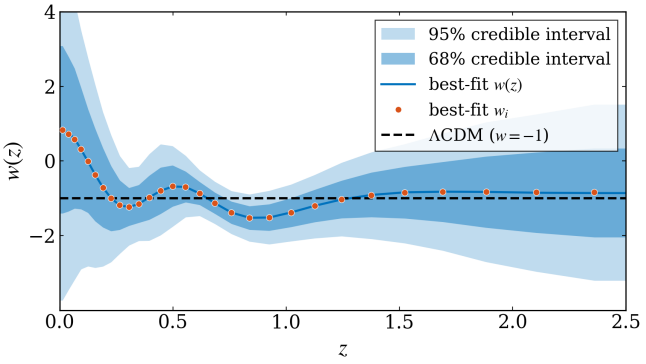


FIG. 1. Fiducial nonparametric dark-energy reconstruction adopted from DESI DR2. Red points denote the best-fitting bin values w_i, while the solid blue curve shows the corresponding reconstructed $w(z)$. The shaded blue regions indicate the 68% and 95% credible intervals, and the horizontal dashed black line marks the ΛCDM prediction, $w = -1$.

$H(z)$ from Eqs. (8) and (9). The DESI-reconstructed expansion history then predicts the fiducial luminosity distance $d_L^{\rm fid}(z)$ used to generate the mock catalogs.

Following Ref. [19], we model the redshift distribution of GW sources as

$$P(z) \propto \frac{4\pi d_C^2(z) R(z)}{H(z)(1+z)}, \tag{19}$$

where the compact-binary merger rate $R(z)$ is [29]

$$R(z) = \begin{cases} 1+2z, & z \le 1, \\ \frac{3}{4}(5-z), & 1 < z < 5, \\ 0, & z \ge 5 \end{cases}. \tag{20}$$

We model each luminosity-distance error using the standard ET forecasting prescription. Because the GW amplitude scales as d_L^{-1}, the Fisher-matrix instrumental error scales as d_L/ρ, where ρ is the ET network signal-to-noise ratio (SNR). We account conservatively for the distance–inclination degeneracy by adopting [31]

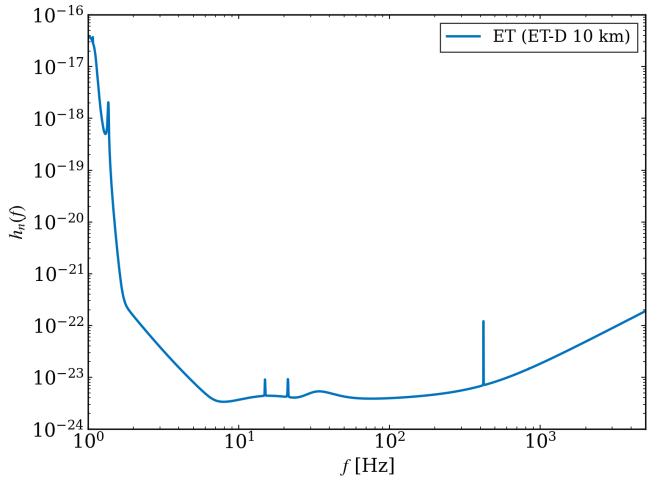


FIG. 2. Characteristic noise strain for the adopted ET-D sensitivity [30].

$$\sigma_{d_L}^{\rm inst} \simeq \frac{2 d_L}{\rho}. \tag{21}$$

The network SNR combines those of the three interferometers in quadrature,

$$\rho = \left[\sum_{j=1}^{3} \left(\rho^{(j)} \right)^2 \right]^{1/2}, \tag{22}$$

where $\rho^{(j)} = \sqrt{\langle \mathcal{H}^{(j)}, \mathcal{H}^{(j)} \rangle}$, denotes the matched-filter SNR of the jth interferometer. The noise-weighted inner product is defined as

$$\langle a, b \rangle = 4 \int_{f_{\rm lower}}^{f_{\rm upper}} \frac{\tilde{a}(f)\tilde{b}^*(f) + \tilde{a}^*(f)\tilde{b}(f)}{2 S_h(f)} \, {\rm d}f, \tag{23}$$

where $\tilde{a}(f)$ and $\tilde{b}(f)$ are Fourier transforms and $S_h(f)$ is the one-sided noise power spectral density. The characteristic noise strain is $h_n(f) = \sqrt{f S_h(f)}$. We use the ET-D noise power spectral density from Ref. [20], shown in Fig. 2.

For each event, we draw the component masses independently from a Gaussian distribution with mean $1.33\,M_\odot$ and standard deviation $0.09\,M_\odot$, truncated to $[1.0, 2.0]\,M_\odot$. This distribution is consistent with the observed Galactic double-neutron-star population [32, 33]. We sample the sky position (θ, ϕ), polarization angle ψ, and inclination ι isotropically. The SNR integral extends from $f_{\rm lower} = 1\,{\rm Hz}$ to $f_{\rm upper} = 2 f_{\rm LSO}$, where $f_{\rm LSO} = 1/(6^{3/2} 2\pi M_{\rm obs})$ and $M_{\rm obs} = (1+z)M$ is the detector-frame total mass. We retain events with

$\rho_{\rm net} > 8$ [19, 31]. We also include the weak-lensing error $\sigma^{\rm lens}_{d_L} = 0.05\, z\, d_L$ [31, 41]. The total luminosity-distance uncertainty is

$$\sigma_{d_L} = \sqrt{(\sigma^{\rm inst}_{d_L})^2 + (\sigma^{\rm lens}_{d_L})^2}. \tag{24}$$

For each source, we draw a mock GW luminosity distance from $d^{\rm mock}_{L,i} \sim \mathcal{N}\left(d^{\rm fid}_{L,i}, \sigma^2_{d_L,i}\right)$. We generate catalogs of $N = 100$, 500, 1000, and 3000 events with identified electromagnetic redshifts. Beyond the network-SNR threshold, we impose no redshift-, inclination-, or sky-dependent electromagnetic selection. We condition on a secure redshift and do not predict an electromagnetic selection function or bright-siren yield [40]. Thus, the catalog sizes count usable bright standard sirens and do not correspond to fixed observing times. Figure 3 shows the resulting $\{z_i, d^{\rm mock}_{L,i}, \sigma_{d_L,i}\}$.

V. EXTERNAL DATA

In this work, for some cases of analysis, we combine each mock GW catalog with the following external data.

A. DESI DR2 BAO

We use 13 DESI DR2 BAO distance measurements spanning $0.295 \le z_{\rm eff} \le 2.33$ [5]. They use four tracer types: the bright galaxy sample (BGS) [48], luminous red galaxies (LRGs) [49], emission-line galaxies (ELGs) [50], and quasars (QSO) [51]. The data vector contains the isotropic BGS measurement $D_V/r_{\rm d}$ at $z_{\rm eff} = 0.295$ and six anisotropic pairs $(D_M/r_{\rm d}, D_H/r_{\rm d})$ from LRG1, LRG2, LRG3+ELG1, ELG2, QSO, and Lyα (Table IV of Ref. [5]). The adopted measurements and their correlation coefficients are summarized in Table II.

B. BBN and CMB

We impose a Gaussian BBN prior on the physical baryon density,

$$\Omega_b h^2 = 0.02196 \pm 0.00063, \tag{25}$$

following Ref. [27].

For the CMB, we use the compressed Planck PR4 acoustic-scale constraint

$$100\theta_* = 1.04098 \pm 0.00042, \tag{26}$$

where

$$\theta_* \equiv \frac{r_*}{D_M(z_*)}. \tag{27}$$

Here r_* is the comoving sound horizon at photon decoupling, z_* is the decoupling redshift, and $D_M(z_*)$ is the transverse comoving distance to last scattering. Following Ref. [7], we set $\sigma(100\theta_*) = 0.00042$, a 75% enlargement of the `PR4_12.6 TTTEEE` uncertainty in Table 5 of Ref. [28].

We assume standard ΛCDM physics before recombination and at the baryon-drag epoch. We compute $r_{\rm d}$, r_*, and z_* self-consistently with `CAMB` [47].

VI. COSMOLOGICAL INFERENCE AND RECONSTRUCTION METHODS

In this section, we first define the common likelihood and sampling procedure, and then describe the nonparametric reconstruction, PCA, effective dimensionality, and complementary CPL model comparison.

A. Cosmological inference

We infer the cosmological parameters with the Metropolis–Hastings Markov chain Monte Carlo (MCMC) sampler implemented in `Cobaya` [34]. For a simulated standard-siren catalog $\{z_i, d^{\rm mock}_{L,i}, \sigma_{d_L,i}\}_{i=1}^N$, the theoretical luminosity distance is evaluated at the source redshifts, $d^{\rm th}_L(z_i;\boldsymbol{\theta})$. In analysis, we assume the redshifts as known and assume independent Gaussian distance errors. Therefore, the likelihood is $\mathcal{L} \propto \exp(-\chi^2/2)$, where

$$\chi^2_{\rm GW}(\boldsymbol{\theta}) = \sum_{i=1}^{N} \frac{\left[d^{\rm mock}_{L,i} - d^{\rm th}_L(z_i;\boldsymbol{\theta})\right]^2}{\sigma^2_{d_L,i}}. \tag{28}$$

When external probes are included, the corresponding DESI BAO, BBN, and CMB contributions are added to $\chi^2_{\rm GW}$ to construct the total data χ^2, using the observables and uncertainties described in Sec. V. Table III lists the uniform parameter priors. A parameter common to multiple models has the same prior in each model.

We require the chains to satisfy the Gelman–Rubin criterion $R - 1 < 0.01$ [35] and verify an effective sample size greater than 10^3 for every sampled parameter. Marginalized posterior summaries and credible regions are obtained with `GetDist` [36].

B. Nonparametric reconstruction of $w(z)$

Our primary analysis reconstructs $w(z)$ without imposing a prescribed two-parameter redshift dependence. We adopt the same representation used to generate the fiducial expansion history in Sec. IV: $w(z)$ is piecewise constant in 29 bins, uniformly spaced in scale factor over $0 < z \le 2.5$. The GW-only analysis samples $\{H_0, \Omega_m, w_1, \ldots, w_{29}\}$, while the combined DESI+BBN+CMB+GW analysis additionally includes $\Omega_b h^2$ as a free parameter.

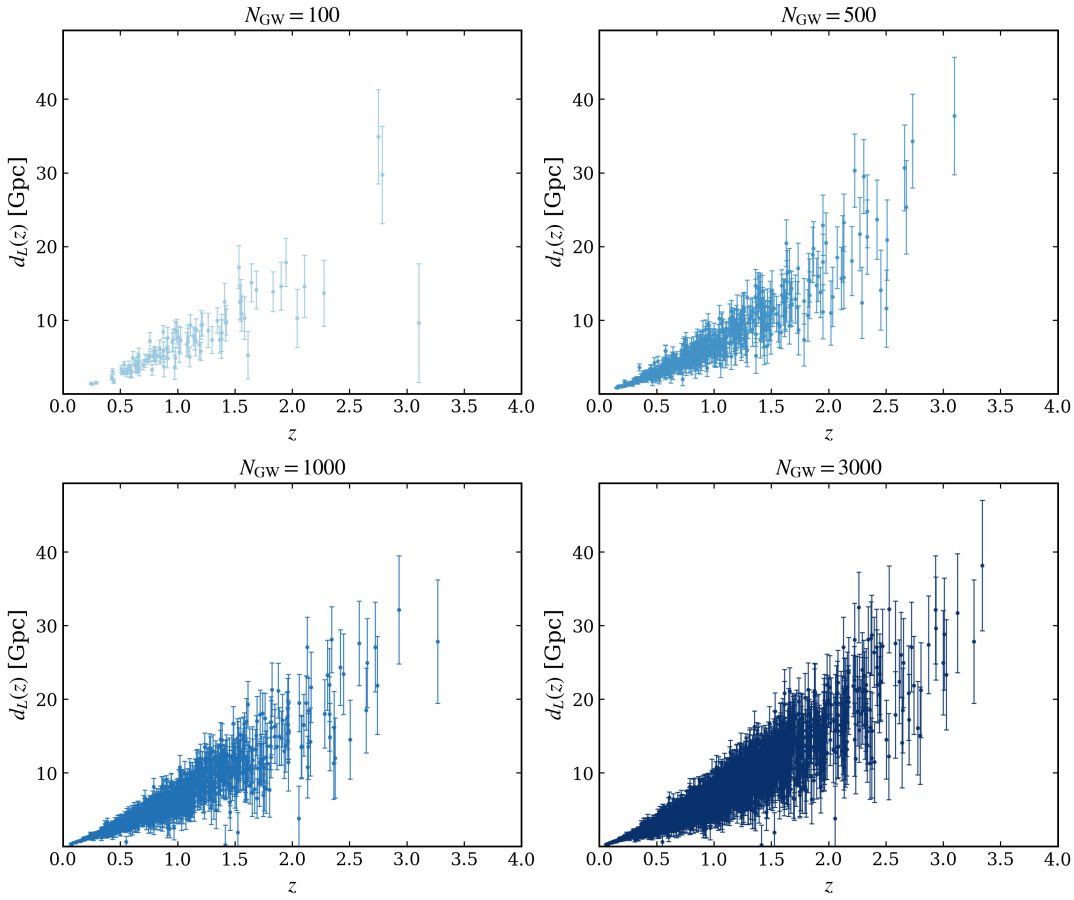


FIG. 3. Mock ET standard-siren catalogs generated from the fiducial DESI DR2 reconstruction for $N = 100$, 500, 1000, and 3000. Each point represents a detected BNS with a redshift, luminosity distance, and total distance uncertainty. The requirement $\rho_{\rm net} > 8$ truncates the catalogs at high redshift.

TABLE II. DESI DR2 BAO distance measurements adopted in this work. The data consist of one isotropic measurement, D_V/r_d, from BGS and six anisotropic measurement pairs, $(D_M/r_d, D_H/r_d)$, from LRG, ELG, QSO, and Lyα tracers [5]).

Tracer	$z_{\rm eff}$	D_V/r_d	D_M/r_d	D_H/r_d
BGS	0.295	7.942 ± 0.075	–	–
LRG1	0.510	–	13.588 ± 0.167	21.863 ± 0.425
LRG2	0.706	–	17.351 ± 0.177	19.455 ± 0.330
LRG3+ELG1	0.934	–	21.576 ± 0.152	17.641 ± 0.193
ELG2	1.321	–	27.601 ± 0.318	14.176 ± 0.221
QSO	1.484	–	30.512 ± 0.760	12.817 ± 0.516
Lyα	2.330	–	38.988 ± 0.531	8.632 ± 0.101

Because the distance observables constrain broad redshift integrals of $w(z)$, the individual bin amplitudes are highly degenerate. We regularize rapid bin-to-bin variations with the Horndeski-motivated correlation covariance $\mathbf{C}_\Pi$ [8]. Its continuous form is $C(a,a') = \sqrt{C(a)C(a')}R(a,a')$, with $C(a) = 0.05 + 0.8a^2$ and $R(a,a') = \exp[-(|\ln a - \ln a'|/0.3)^{1.2}]$. Following Refs. [7, 45, 46], we define a five-bin moving-average matrix $\mathbf{S}$, with its weights renormalized near the boundaries, and penalize departures from the locally smoothed history through

$$\chi^2_{\rm prior} = (\mathbf{w} - \mathbf{S}\mathbf{w})^T \mathbf{C}_\Pi^{-1} (\mathbf{w} - \mathbf{S}\mathbf{w}). \tag{29}$$

Equivalently, the prior precision in the w-bin space is $\widetilde{\mathbf{C}}_\Pi^{-1} = (\mathbf{I} - \mathbf{S})^T \mathbf{C}_\Pi^{-1} (\mathbf{I} - \mathbf{S})$.

For the 32-dimensional parameter vector within the prior ranges of Table III, the posterior is sampled using two types of catalogs with the correla-

TABLE III. Uniform parameter priors. The nonparametric analysis varies H_0, Ω_m, $\Omega_b h^2$, and the 29 binned equation-of-state amplitudes w_i. The GW-only CPL analysis varies H_0, Ω_m, w_0, and w_a. In flat ΛCDM, $w_0 = -1$ and $w_a = 0$ are fixed.

Parameter	Prior
H_0 [km s^{-1} Mpc^{-1}]	$[20, 100]$
Ω_m	$[0.01, 0.99]$
w_0	$[-3, 3]$
w_a	$[-30, 3]$
$\Omega_b h^2$	$[0.005, 0.04]$
$w_i (i = 1, \ldots, 29)$	$[-6, 4]$

tion prior: (i) GW-only catalogs, and (ii) combined DESI+BBN+CMB+GW catalogs.

$$\chi^2 = \chi^2_{\rm data} + A\chi^2_{\rm prior}. \tag{30}$$

The factor A controls the strength of the correlation prior relative to the data. Here, we adopt $A = 1$ for the reconstructed $w(z)$ history. The results are shown in Fig. 4. The GW-only catalogs provide limited constraints on the reconstructed $w(z)$ history due to the intrinsic distance-redshift degeneracy. Even for the largest catalog with 3000 standard sirens, the input $w(z)$ history can only be reliably recovered at low redshifts ($z \lesssim 0.5$), while the reconstruction remains poorly constrained at higher redshifts. In contrast, the inclusion of DESI DR2 BAO, BBN, and the compressed CMB constraint significantly improves the reconstruction capability, leading to substantially tighter constraints over the low- and intermediate-redshift ranges. All combined DESI+BBN+CMB+GW analyses exhibit a consistent evolutionary pattern in $w(z)$, with $w > -1$ at $z \lesssim 0.2$ and $w < -1$ around $z \simeq 0.7$–1.0, closely following the injected dark-energy history.

C. Principal components analysis

The w bins are strongly correlated, so the data constrain combinations of bins rather than each bin independently. We use PCA to identify these modes [53],

$$\mathbf{F}_w \equiv \mathbf{C}_w^{-1} = \mathbf{E}^T \mathbf{\Lambda} \mathbf{E}, \quad \mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_{N_w}), \tag{31}$$

where $N_w = 29$, the rows of $\mathbf{E}$ are the orthonormal eigenmodes, and larger λ_i correspond to more tightly constrained modes. For each catalog, $\mathbf{C}_w$ is the covariance matrix of the 29 w bins for the $A = 1$ reconstruction, after marginalizing over H_0, Ω_m, and $\Omega_b h^2$.

We identify $\mathbf{w}_{\rm best}$ with the 29-component w-bin vector from the converged MCMC sample having the minimum total effective χ^2. Relative to the ΛCDM reference history, $\mathbf{w}_\Lambda = (-1, \ldots, -1)^T$, the reconstructed departure is expanded as

$$\mathbf{w}_{\rm best} - \mathbf{w}_\Lambda = \sum_{i=1}^{N_w} \gamma_i \mathbf{e}_i, \quad \gamma_i = \mathbf{e}_i^T \left(\mathbf{w}_{\rm best} - \mathbf{w}_\Lambda\right). \tag{32}$$

The marginalized uncertainty of the ith mode is $\sigma(\gamma_i) = \lambda_i^{-1/2}$, and its signed signal-to-noise ratio is $s_i \equiv \gamma_i/\sigma(\gamma_i) = \gamma_i\sqrt{\lambda_i}$. The cumulative reconstruction significance (that is, the signal-to-noise ratio, SNR) over the first m modes is

$$\mathrm{SNR}(m) = \left[\sum_{i=1}^{m} s_i^2\right]^{1/2}. \tag{33}$$

This cumulative SNR measures the overall significance of the departure from the reference history contained in the first m PCA modes.

We further evaluate the Bayesian evidence ratio between the PCA-truncated nonparametric $w(z)$ model and ΛCDM. For a model retaining the first m principal components, the Bayes factor is defined as

$$\ln B_{m,\Lambda} = \ln Z_m - \ln Z_{\Lambda\rm CDM}, \tag{34}$$

where Z_m and $Z_{\Lambda\rm CDM}$ denote the Bayesian evidences of the m-mode $w(z)$ model and the ΛCDM model, respectively. By varying the number of retained modes, we assess how the Bayesian preference for dynamical dark energy depends on the effective dimensionality of the reconstruction. The Bayesian evidence is computed using the nested-sampling algorithm implemented in `dynesty` [54].

Table IV summarizes the covariance-weighted reconstruction significance and Bayesian evidence obtained from the first m principal components for both the GW-only and the combined DESI+BBN+CMB+GW analyses.

For both analyses, the deviation from ΛCDM is mainly captured by the first few principal components, indicating that the dynamical features of the injected $w(z)$ history are dominated by a small number of correlated modes. For the GW-only catalogs, the constraining power increases rapidly with the number of standard sirens. The GW3000 catalog reaches a covariance-weighted significance of SNR = 7.170 and a Bayesian evidence ratio of $\ln B_{7,\Lambda} = 20.874$, while the GW1000 catalog provides a moderate preference with SNR = 3.299 and $\ln B_{7,\Lambda} = 3.563$. This demonstrates that a sufficiently large bright-siren catalog can independently constrain departures from ΛCDM, although the distance–redshift information alone provides limited sensitivity for smaller catalogs.

The inclusion of DESI DR2 BAO, BBN, and CMB constraints substantially enhances the sensitivity to the reconstructed dark-energy history. For the GW500 catalog, retaining the first six principal components gives SNR = 4.351 with $\ln B_{6,\Lambda} = 5.048$. For catalogs reaching $\mathcal{O}(10^3)$ events, the covariance-weighted significance exceeds 5σ and the Bayesian evidence strongly favors the

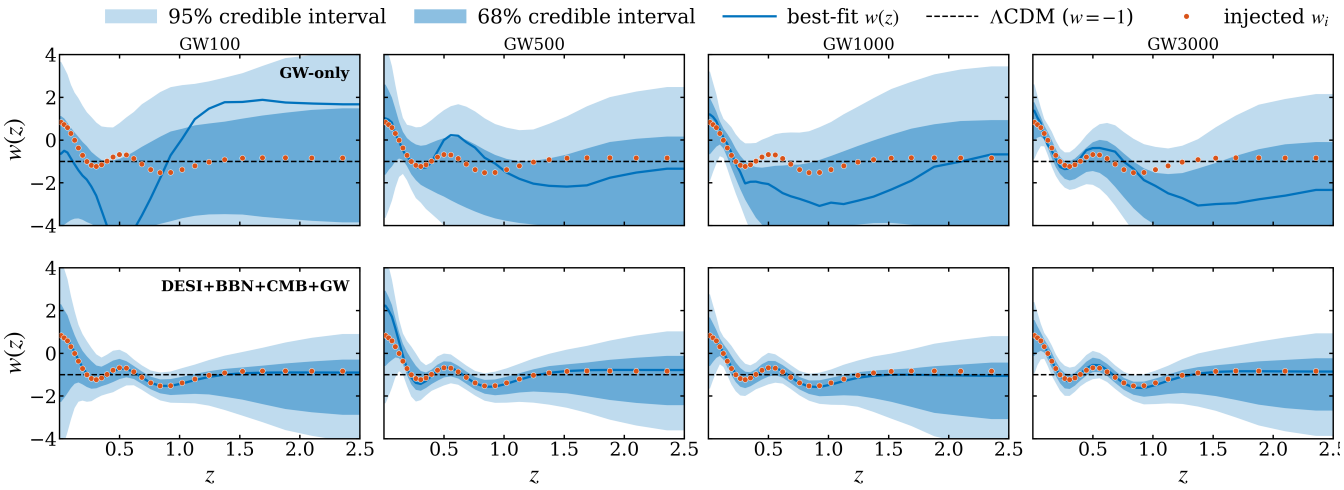


FIG. 4. Nonparametric reconstructions of the dark-energy equation of state obtained from the GW-only and the combined DESI+BBN+CMB+GW analyses for mock ET catalogs containing 100, 500, 1000, and 3000 standard sirens. The dark- and light-blue bands show the 68% and 95% credible intervals, respectively; the solid blue curves show the best-fitting reconstructions, the orange points mark the injected bin values, and the dashed black lines denote the flat-ΛCDM prediction $w = -1$.

TABLE IV. Comparison of covariance-weighted reconstruction significance and Bayesian evidence ratios for two analyses. The quantities SNR and $\ln B_{m,\Lambda}$ are calculated using the first m principal components.

GW-only

	SNR				$\ln B_{m,\Lambda}$			
m	GW100	GW500	GW1000	GW3000	GW100	GW500	GW1000	GW3000
1	0.058	1.364	3.194	6.844	-0.003	-0.258	3.859	21.386
2	0.103	1.399	3.209	7.073	-0.007	-0.346	3.550	21.445
3	0.108	1.442	3.279	7.109	-0.009	-0.398	3.701	20.965
4	0.172	1.444	3.296	7.129	-0.020	-0.435	3.712	20.801
5	0.173	1.444	3.298	7.161	-0.022	-0.438	3.624	20.947
6	0.191	1.493	3.298	7.167	-0.026	-0.450	3.621	20.894
7	0.193	1.510	3.299	7.170	-0.028	-0.488	3.563	20.874

DESI+BBN+CMB+GW

	SNR				$\ln B_{m,\Lambda}$			
m	GW100	GW500	GW1000	GW3000	GW100	GW500	GW1000	GW3000
1	1.062	2.856	5.854	9.402	-1.507	1.669	14.267	40.880
2	1.965	3.823	7.577	11.072	-0.684	4.211	24.582	56.315
3	2.053	3.824	7.696	11.169	-0.651	3.958	24.830	56.780
4	2.079	3.961	7.868	11.365	-0.752	4.279	25.975	58.716
5	2.097	4.297	7.880	11.373	-0.779	4.997	25.870	58.596
6	2.155	4.351	7.913	11.412	-1.283	5.048	25.851	58.351
7	2.173	4.360	7.937	11.412	-1.351	4.372	25.296	58.049

dynamical dark-energy reconstruction when several degrees of freedom are retained. In particular, for $m = 4$, the Bayesian evidence reaches its maximum for both GW1000 and GW3000, with $\ln B_{4,\Lambda} = 25.975$ and 58.716, corresponding to SNR = 7.868 and 11.365, respectively.

To isolate the modes constrained by the data, we perform an auxiliary MCMC reconstruction with $A = 0.01$ in Eq. (30) and denote the resulting data-posterior covariance of $\mathbf{w}_\mathrm{d}$ by $\mathbf{C}_\mathrm{d}$. The data precision matrix is then

estimated by subtracting the residual prior contribution,

$$\mathbf{F}_{\rm data} = \mathbf{C}_{\rm d}^{-1} - 0.01\widetilde{\mathbf{C}}_{\Pi}^{-1}. \tag{35}$$

We apply Eq. (31) separately to $\mathbf{F}_{\rm data}$ and $\widetilde{\mathbf{C}}_{\Pi}^{-1}$ to obtain the eigenvalues.

Fig. 5 shows the principal-component precision spectra of the reconstructed dark-energy equation of state for the GW-only and combined DESI+BBN+CMB+GW analyses. The eigenvalues λ_i quantify the constraining power of each principal component. For both analyses, the information is predominantly concentrated in the first few principal components. Increasing the number of ET standard sirens enhances the leading eigenvalues, indicating that larger GW catalogs mainly improve the precision of the best-constrained correlated combinations of the $w(z)$ history. However, the eigenvalues remain small for most higher-order modes in the GW-only case, reflecting the limited ability of luminosity-distance measurements alone to resolve the full 29-bin $w(z)$ parameter space. The inclusion of DESI DR2 BAO, BBN, and CMB constraints significantly increases the leading eigenvalues by providing additional information on the expansion history. The apparent truncation near the 25th mode arises from several poorly constrained directions with very small eigenvalues, corresponding to rapidly varying features between neighboring bins that are suppressed by the Horndeski correlation prior rather than independently measured by the data.

The corresponding eigenmodes in Fig. 6 illustrate the redshift dependence of the correlated combinations constrained by the data. In both the GW-only and combined analyses, the first principal component (PC1) exhibits a broad and smooth structure, indicating that the best-constrained combination mainly probes the overall variation of the dark-energy history over a wide redshift range. Higher-order modes (PC2–PC4) contain additional nodes and oscillatory features, corresponding to more localized redshift-dependent variations that are progressively less constrained.

For the GW-only analysis, the shapes of the leading eigenmodes become more stable as the catalog size increases from 100 to 3000 events. In particular, the small-scale oscillatory features present in the poorly constrained higher-order modes of the GW100 case are gradually reduced, while the dominant structures of PC1–PC4 are preserved. When DESI DR2 BAO, BBN, and CMB constraints are included, the eigenmodes are more localized and better determined, especially for the higher-order components. The external cosmological probes provide complementary constraints on the expansion history and reduce the degeneracies among different redshift bins. Nevertheless, the overall structure of the leading principal components remains similar, suggesting that the additional information primarily sharpens the measurement of the same dominant $w(z)$ combinations.

Following Ref. [52], we characterize the effective number of independent $w(z)$ modes constrained beyond the correlation prior by

$$N_{\rm eff} = 29 - {\rm Tr}\left(\widetilde{\mathbf{C}}_{\Pi}^{-1}\mathbf{C}_w\right). \tag{36}$$

For the GW-only analysis, $N_{\rm eff}$ increases from 3.828 for GW100 to 5.191 for GW3000. With the inclusion of DESI DR2 BAO, BBN, and the compressed CMB acoustic-scale constraint, the effective dimensionality becomes $N_{\rm eff} = 5.363$, 5.599, 5.888, and 6.274 for GW100, GW500, GW1000, and GW3000, respectively. Although the external probes substantially enhance the constraining power, the effective dimensionality remains far below the original 29-dimensional parameter space. Therefore, the reconstructed dark-energy history is effectively described by only a small number of correlated modes, with increasing GW statistics mainly improving their precision rather than opening many new independent degrees of freedom.

D. Parametric CPL analysis

As a complementary low-dimensional diagnostic, we fit each GW catalog under spatially flat ΛCDM and its Chevallier–Polarski–Linder (CPL) extension. Because the injected dark-energy history does not follow the CPL form, the inferred (w_0, w_a) values should be interpreted as an effective projection of its luminosity-distance relation, rather than as a direct recovery of the input $w(z)$. We consider two types of catalogs: (i) GW-only catalogs, where constraints are derived solely from standard siren measurements, and (ii) combined DESI+BBN+CMB+GW catalogs, where external cosmological probes are included. The sampled parameter vectors of GW-only are

$$\boldsymbol{\theta}_{\Lambda\rm CDM} = (H_0, \Omega_m), \quad \boldsymbol{\theta}_{\rm CPL} = (H_0, \Omega_m, w_0, w_a), \tag{37}$$

while the combined DESI+BBN+CMB+GW analysis additionally includes $\Omega_b h^2$ as a free parameter. The priors listed in Table III. These fits use the GW likelihood in Eq. (28); one-dimensional constraints are reported as marginalized posterior means and standard deviations.

We refine the maximum-likelihood estimates (MLEs) with `iminuit` [37], initializing the optimizer from the highest-posterior samples of the independent chains. Among the converged solutions, we select the one with the minimum χ^2. The improvement over ΛCDM is quantified by

$$\Delta\chi^2 = \chi^2(\hat{\boldsymbol{\theta}}_{\Lambda\rm CDM}) - \chi^2(\hat{\boldsymbol{\theta}}_{\rm CPL}), \tag{38}$$

where hats denote MLEs. Thus, $\Delta\chi^2 > 0$ indicates that CPL gives the better maximum-likelihood fit.

Flat ΛCDM is nested within CPL at $(w_0, w_a) = (-1, 0)$, and the two models differ by two free parameters. Under the regularity conditions of Wilks' theorem [38], $\Delta\chi^2$ follows a χ^2 distribution with two degrees of freedom under the ΛCDM null hypothesis. We quote

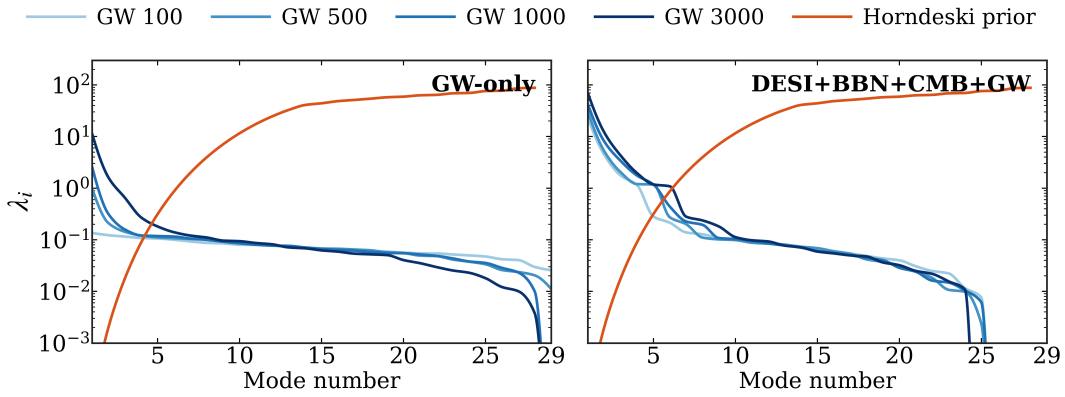


FIG. 5. Principal-component precision spectra for the nonparametric dark-energy reconstructions obtained from the GW-only and the combined DESI+BBN+CMB+GW analyses. The blue curves show the data eigenvalue spectra for different numbers of GW events, while the orange curve shows the eigenvalue spectrum of the Horndeski correlation prior.

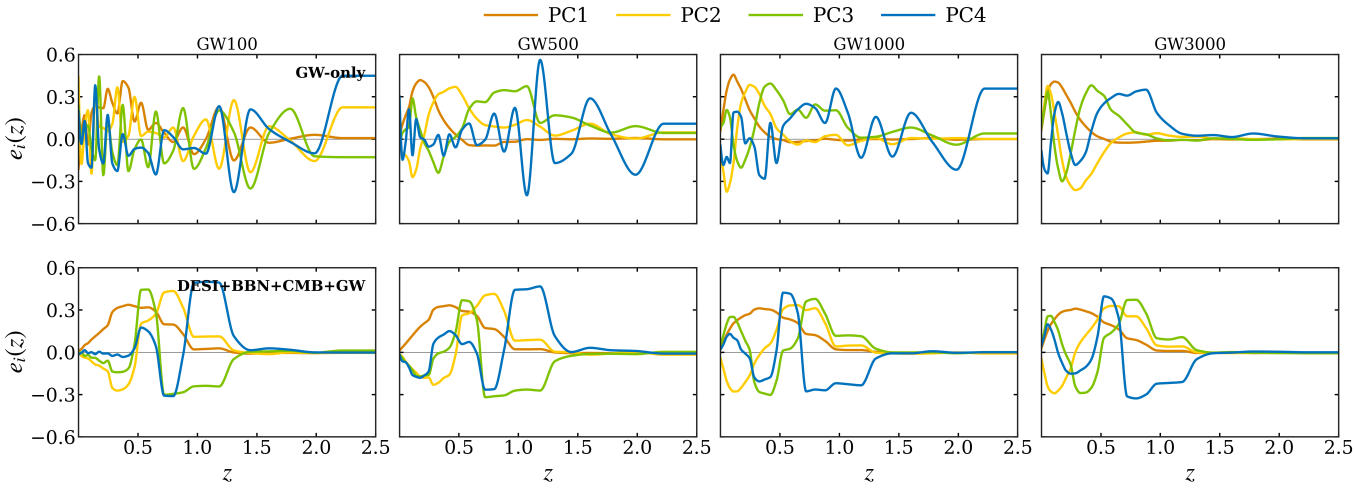


FIG. 6. The first four principal component eigenmodes for the same combinations of the GW-only and the combined DESI+BBN+CMB+GW analyses. The four panels correspond to 100, 500, 1000, and 3000 GW events.

a two-sided Gaussian-equivalent significance N_σ defined by

$$\mathrm{CDF}_{\chi^2}\left(\Delta\chi^2 \mid 2\,\mathrm{dof}\right) = \frac{1}{\sqrt{2\pi}} \int_{-N_\sigma}^{N_\sigma} e^{-t^2/2}\,\mathrm{d}t. \qquad (39)$$

Figure 7 shows the marginalized CPL constraints in the (w_0, w_a) plane for the four representative ET standard-siren catalogs. The blue contours correspond to the GW-only likelihood, while the orange contours include the combined DESI+BBN+CMB+GW constraints.

For the GW-only analysis, the constraints exhibit a strong anticorrelation between w_0 and w_a, reflecting the degeneracy of the CPL parameterization. For 100 events, the broad posterior contains the ΛCDM point $(w_0, w_a) = (-1, 0)$. With 500 standard sirens, the allowed region becomes narrower, although ΛCDM remains compatible with the 95% credible region. For the 1000- and 3000-event catalogs, the posterior shifts away from ΛCDM and is concentrated in the Quintom-B sector, defined by $w_0 > -1$ and $w_0 + w_a < -1$ [39]. These CPL parameters should be interpreted as an effective low-dimensional description of the injected binned equation-of-state history rather than the original $w(z)$ bins themselves. The inclusion of DESI+BBN+CMB data further tightens the CPL constraints and enhances the displacement from ΛCDM. This improvement reflects the complementary information provided by early- and late-universe probes when combined with standard sirens.

Table V quantifies this displacement using the Wilks likelihood-ratio statistic. For the GW-only case, the 100- and 500-event CPL maximum-likelihood solutions reach the prior boundaries, and therefore the corresponding Wilks significances are not reported. For the 1000- and 3000-event catalogs, the nominal significances are 3.309σ and 6.317σ, respectively. When DESI+BBN+CMB con-

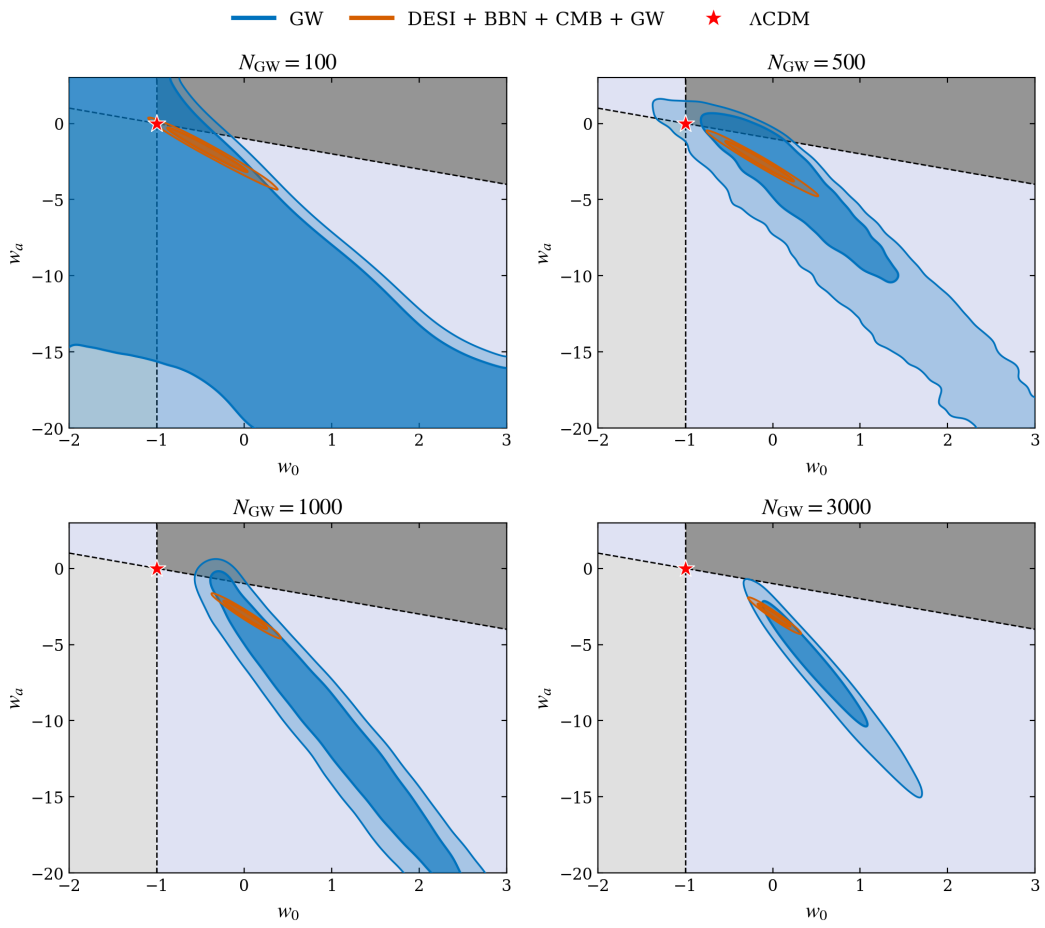


FIG. 7. Marginalized 68% and 95% credible regions in the (w_0, w_a) plane from CPL fits to four representative ET standard-siren catalogs with $N_{\rm GW} = 100, 500, 1000$, and 3000. The blue contours show constraints from the GW-only likelihood, while the orange contours correspond to the combined DESI DR2 DESI+BBN+CMB+GW analysis. The red star marks the flat ΛCDM point, $(w_0, w_a) = (-1, 0)$. The dashed lines indicate $w_0 = -1$ and $w_0 + w_a = -1$, which separate the different CPL sectors. The fitted CPL parameters provide an effective description of the injected binned dynamical dark-energy history.

TABLE V. Comparison of the Wilks-theorem model-comparison statistics for the GW-only and DESI+BBN+CMB+GW analyses. The significance N_σ is obtained from $\Delta\chi^2$ assuming two additional degrees of freedom ($\Delta k = 2$). For the GW-only case, the CPL maximum-likelihood solutions for 100 and 500 events reach the prior boundaries, and therefore the corresponding significances are not reported.

	GW-only		DESI+BBN+CMB+GW	
$N_{\rm GW}$	$\Delta\chi^2$	N_σ	$\Delta\chi^2$	N_σ
100	0.629	–	4.348	1.582
500	7.950	–	16.694	3.676
1000	13.944	3.309	64.010	7.710
3000	44.086	6.317	127.004	11.033

straints are included, the corresponding values increase to 7.710σ and 11.033σ. These results demonstrate that, for the injected dynamical-dark-energy history and the adopted mock realizations, future ET standard sirens combined with external cosmological probes can substantially improve the discrimination between flat ΛCDM and its effective CPL extension.

VII. CONCLUSIONS

We have investigated the capability of Einstein Telescope (ET) bright standard sirens to test a dynamical-dark-energy expansion history motivated by DESI DR2. We adopted the best-fitting 29-bin reconstruction of $w(z)$

from Ref. [7] as a fiducial cosmology and generated mock binary-neutron-star bright standard-siren catalogs with identified host redshifts. The analysis was performed in two complementary frameworks: a nonparametric reconstruction of the dark-energy history using the 29-bin $w(z)$ parameterization, and a parametric comparison between flat ΛCDM and the CPL extension using the GW likelihood.

The nonparametric analysis demonstrates that ET standard sirens constrain a small number of correlated redshift modes rather than individual $w(z)$ bins. The PCA analysis shows that the leading principal components contain most of the information in both the GW-only and the combined DESI+BBN+CMB+GW analyses. Increasing the number of standard sirens mainly improves the precision of these already constrained modes, while the overall structures of the leading eigenmodes remain stable. For the GW-only analysis, the effective dimensionality increases from $N_{\rm eff} = 3.828$ for GW100 to 5.191 for GW3000. When DESI DR2 BAO, BBN, and the compressed CMB acoustic-scale constraint are included, the corresponding values become $N_{\rm eff} = 5.363$, 5.599, 5.888, and 6.274 for GW100, GW500, GW1000, and GW3000, respectively. Therefore, even with thousands of standard sirens, the reconstructed 29-dimensional dark-energy history is effectively described by only a small number of correlated modes.

The covariance-weighted reconstruction significance and Bayesian evidence further quantify the departure from ΛCDM. For the GW-only catalogs, the constraining power increases rapidly with the number of standard sirens. For the GW3000 catalog, the cumulative reconstruction significance exceeds 5σ and the Bayesian evidence satisfies $\ln B_{m,\Lambda} > 10$ for all retained modes from $m = 1$ to 7, indicating a strong preference for the dynamical-dark-energy reconstruction over ΛCDM. In contrast, smaller catalogs provide weaker statistical evidence due to their limited distance–redshift information. The inclusion of DESI DR2 BAO, BBN, and CMB constraints significantly enhances the sensitivity. For example, the GW500 catalog reaches SNR = 4.351 and $\ln B_{6,\Lambda} = 5.048$ when the first six principal components are retained, and the GW1000 and GW3000 catalogs achieve SNR = 7.868 and 11.365, respectively, with the maximum Bayesian evidence obtained at four retained modes. These results indicate that the deviation from ΛCDM induced by the fiducial dynamical-dark-energy history is primarily encoded in a few broad correlated modes.

The GW-only CPL analysis provides a complementary low-dimensional description of the same expansion history. Since the injected $w(z)$ evolution is not restricted to the CPL form, the recovered (w_0, w_a) parameters should be interpreted as an effective projection of the luminosity-distance relation rather than a direct reconstruction of the original 29-bin history. For 100 and 500 standard sirens, the CPL maximum-likelihood solutions reach the prior boundaries, preventing a reliable Wilks-theorem significance conversion. For 1000 and 3000 events, the nominal Gaussian-equivalent significances are 3.31σ and 6.32σ, respectively, under the two-degree-of-freedom Wilks approximation.

When DESI BAO, BBN, and CMB acoustic-scale constraints are combined with the GW catalogs, the CPL constraints become substantially tighter, with nominal significances increasing to 7.71σ and 11.03σ for the 1000- and 3000-event catalogs. The corresponding CPL parameters remain in the Quintom-B sector, characterized by $w_0 > -1$ and $w_0 + w_a < -1$, showing that this effective two-parameter description captures the crossing-like behavior of the injected dynamical-dark-energy history.

Under the assumptions of the adopted DESI DR2-motivated fiducial history, the idealized bright-standard-siren catalogs, and the forecast ET distance precision, these results demonstrate that future ET observations can provide an independent probe of departures from the flat ΛCDM expansion history. Combining standard sirens with complementary cosmological probes will be essential for maximizing their capability to test dynamical dark energy.

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